Teorema reziduurilor - exerciții

ex $I = \int_{\gamma} \frac{\operatorname{tg} z}{z^2(z^2+1)}\,dz$

$\gamma: 0{,}25x^2 + 0{,}16y^2 = 0{,}04$

$\gamma: \frac{x^2}{\frac{100}{25}} + \frac{y^2}{\frac{100}{16}} = \frac{4}{100}$

$\gamma: \frac{x^2}{\left(\frac{2}{5}\right)^2} + \frac{y^2}{\left(\frac{1}{2}\right)^2} = 1$

elipsă de semiaxe $\frac{2}{5}$ și $\frac{1}{2}$

$z=0$ pol de ordinul 1 $\in \operatorname{Int}\gamma$

$z=\pm i$ poli de ordinul 1 $\notin \operatorname{Int}\gamma$

$\cos z = 0 \Rightarrow z = \frac{\pi}{2} + k\pi$ $k \in \mathbb{Z}$ poli de ordinul 1 $\notin \operatorname{Int}\gamma$

$$\lim_{z\to 0} \frac{\operatorname{tg} z}{z^2(z^2+1)} = \lim_{z\to 0} \underbrace{\frac{\operatorname{tg} z}{z}}_{\to 1} \cdot \underbrace{\frac{1}{z(z^2+1)}}_{\to 1} = \infty$$

$I = 2\pi i \operatorname{Rez}(f, 0)$

$$\operatorname{Rez}(f,0) = \lim_{z\to 0} z \cdot \frac{\operatorname{tg} z}{z^2(z^2+1)} = 1$$

$\Rightarrow I = 2\pi i$

ex $\int_{|z|^2-\mathrm{Re}\,z=0} \frac{z^3 e^{\frac{z}{z-1}}}{(z-1)^2}\,dz$

Caz. gen $e^{\frac{1}{z}}$, $z=0$ pct. sing. esențial

$z_n^1 = \frac{1}{2n\pi i} \to 0$; $e^{\frac{1}{z_n^1}} = e^{2n\pi i} = \cos(2n\pi) + i\sin(2n\pi) = 1$

$z_n^2 = \frac{1}{n} \to 0$; $e^{\frac{1}{z_n^2}} = e^n \to \infty$

$\Rightarrow \nexists \lim_{z\to 0} e^{\frac{1}{z}}$

Conform cazului general $\Rightarrow$ $z=1$ pct. sg. esențial

$\gamma: x^2+y^2-2x=0 \Rightarrow (x-1)^2+y^2=1$

$\Rightarrow z=1 \in \mathrm{Int}\,\gamma$

$J = 2\pi i\, \mathrm{Rez}(f,1)$

$$\frac{z^3 \cdot e^{\frac{z}{z-1}}}{(z-1)^2} = \frac{1}{(z-1)^2}(z-1+1)^3 \cdot e^{\frac{z-1+1}{z-1}}$$

$$= \frac{1}{(z-1)^2}\left[(z-1)^3+3(z-1)^2+3(z-1)+1\right] e\left[1+\frac{1}{z-1}+\frac{1}{(z-1)^2\cdot 2!}+\dots\right]$$

$$\mathrm{Rez}(f,1) = e\left(3+3+\frac{1}{2!}\right) = \frac{13e}{2}$$

$$J = 2\pi i \cdot \frac{13e}{2} = 13e\pi i$$

ex $\int_{|z|=2} \frac{e^{\frac{\pi}{z-i}}}{z^2+1}\,dz$

$z=i$ punct singular esențial $\in Int(|z|=2)$

$z=-i$ pol de ordinul 1

$$I = 2\pi i \left(Rez(f,i) + Rez(f,-i)\right)$$

$$Rez(f,-i) = \lim_{z\to -i} \cancel{(z+i)} \frac{e^{\frac{\pi}{z-i}}}{\cancel{z^2+1}\ z-i} = \frac{e^{\frac{\pi}{-2i}}}{-2i}$$

$$\frac{e^{\frac{\pi}{z-i}}}{z^2+1} = \frac{1}{(z+i)(z-i)} \cdot e^{\frac{\pi}{z-i}} = \frac{1}{z-i} \cdot \frac{1}{z-i+2i} \cdot e^{\frac{\pi}{z-i}} =$$

$$= \frac{1}{z-i} \cdot \frac{1}{2i} \cdot \frac{1}{1+\frac{z-i}{2i}} \cdot e^{\frac{\pi}{z-i}}$$

$$= \frac{1}{2i} \cdot \frac{1}{z-i} \left(\sum_{n\geq 0} (-1)^n \cdot \left(\frac{z-i}{2i}\right)^n\right)\left(\sum_{n\geq 0} \left(\frac{\pi}{z-i}\right)^n \cdot \frac{1}{n!}\right)$$

$$= \frac{1}{2i} \cdot \frac{1}{z-i}\left(1 - \frac{z-i}{2i} + \frac{(z-i)^2}{(2i)^2} - \frac{(z-1)^3}{(2i)^3} + \dots\right)\left(1 + \frac{\pi}{(z-i)1!} + \frac{\pi^2}{(z-i)^2 2!} + \right.$$

$$\left. + \frac{\pi^3}{(z-1)^3 3!} + \dots\right)$$

$$\Rightarrow Rez(f,i) = \frac{1}{2i}\left(1 - \frac{\pi}{(2i)1!} + \frac{\pi^2}{(2i)^2 2!} - \frac{\pi^3}{(2i)^3 3!} + \dots\right) = \frac{1}{2i} e^{-\frac{\pi}{2i}}$$

$$I = 2\pi i\left(\frac{1}{2i} e^{-\frac{\pi}{2i}} - \frac{1}{2i} e^{-\frac{\pi}{2i}}\right) = 0$$

ex $\int_\gamma \frac{z^{13}}{(z-2)^4(z^5+3)^2}\,dz \overset{\text{Not.}}{=} I$ $\quad \gamma: 4x^2+9y^2-36=0$

$\frac{x^2}{9}+\frac{y^2}{4}=1$

elipsă de semiaxe 3 și 2

$z=2$ pol de ordinul 4 $\in \text{Int}\,\gamma$

$z^5+3=0 \Rightarrow z^5=-3 \Rightarrow z_k=\sqrt[5]{-3}=(-3)^{\frac{1}{5}}=e^{\frac{1}{5}\ln(-3)}$

$z_k=\{e^{\frac{1}{5}[\ln|-3|+i(\arg(-3)+2k\pi)]},\ k=\overline{0,4}\}$

$=\{\sqrt[5]{3}\cdot e^{\frac{i}{5}(\pi+2k\pi)},\ k=\overline{0,4}\}$ poli de ordinul 2 $\in \text{Int}\,\gamma$

$I=2\pi i\left(\text{Rez}(f,2)+\sum_{k=0}^{4}\text{Rez}(f,z_k)\right)\overset{\text{Cor}}{=}-2\pi i\,\text{Rez}(f,\infty)$

$\text{Rez}(f,\infty)=\text{Rez}\left(-\frac{1}{t^2}f\left(\frac{1}{t}\right),0\right)$

$-\frac{1}{t^2}f\left(\frac{1}{t}\right)=-\frac{1}{t^2}\cdot\frac{\frac{1}{t^{13}}}{\left(\frac{1}{t}-2\right)^4\left(\frac{1}{t^5}+3\right)^2}=-\frac{1}{t^2}\cdot\frac{t^9\cdot t^{10}}{t^{13}(1-2t)^4(1+3t^5)^2}$

$=-\frac{1}{t}\cdot(1-2t)^{-4}\cdot(1+3t^5)^{-2}$

$=-\frac{1}{t}\left(1+\sum_{u\geq 1}\frac{-4(-4-1)\dots(-4-u+1)}{u!}(-2t)^u\right)\left(1+\sum_{u\geq 1}\frac{-2(-2-1)\dots(-2-u+1)}{u!}\cdot 3t^5\right)$

$\Rightarrow \text{Rez}\left(-\frac{1}{t^2}f\left(\frac{1}{t}\right),0\right)=1$

$I=-2\pi i\cdot 1=-2\pi i$

Calculul unor integrale reale cu ajutorul th. reziduurilor

Tipul I: $I = \int_0^{2\pi} R(\sin t, \cos t)\,dt$, unde R este o funcție rațională, al cărei numitor nu se anulează pe cercul unitate

S.V. $z = e^{it}$ $\quad t \in [0, 2\pi) \Rightarrow |z| = 1$

$$dz = ie^{it}\,dt \Rightarrow dt = \frac{1}{ie^{it}}\,dz$$

$$\sin t = \frac{e^{it} - e^{-it}}{2i} = \frac{z - z^{-1}}{2i} = \frac{z^2 - 1}{2zi}$$

$$\cos t = \frac{e^{it} + e^{-it}}{2} = \frac{z + z^{-1}}{2} = \frac{z^2 + 1}{2z}$$

$$I = \int_0^{2\pi} R(\sin t, \cos t)\,dt \xrightarrow{z = e^{it}} I = \int_{|z|=1} R\left(\frac{z^2-1}{2iz}, \frac{z^2+1}{2z}\right) \cdot \frac{dz}{iz}$$

ex $\int_0^{2\pi} \frac{\cos^2 t}{5 - 3\sin t}\,dt \overset{\text{Not.}}{=} I$

$$I = \int_{|z|=1} \frac{\left(\frac{z^2+1}{2z}\right)^2}{5 - 3 \cdot \frac{z^2-1}{2zi}} \cdot \frac{dz}{iz} = \int_{|z|=1} \frac{z^4 + 2z^2 + 1}{4z^2\left(\frac{10iz - 3z^2 + 3}{2zi}\right)}\,dz$$

$$I = \int_{|z|=1} \frac{z^4+2z^2+1}{2z^2(-3z^2+10iz+3)}\,dz = -\frac{1}{2}\int_{|z|=1} \frac{z^4+2z^2+1}{z^2(3z^2-10iz-3)}\,dz$$

$z=0$ pol de ordinul 2 $\in$ Int $(|z|=1)$

$3z^2-10iz-3=0$

$\Delta = -100+4\cdot 3\cdot 3 = 36-100 = -64 = (i\cdot 8)^2 \longrightarrow z_{1,2} = \frac{10i \pm 8i}{2\cdot 3}$

$z_1 = 3i \notin \text{Int}(|z|=1)$

$z_2 = \frac{i}{3} \in \text{Int}(|z|=1)$

$$I = -\frac{1}{2}\cdot 2\pi i\left(\text{Rez}(f,0) + \text{Rez}(f,\tfrac{i}{3})\right)$$

$$\text{Rez}(f,\tfrac{i}{3}) = \lim_{z\to i/3}\left(z-\tfrac{i}{3}\right)\frac{(z+1)^2}{z^3\cdot 3(z-3i)\left(z-\frac{i}{3}\right)} = \frac{\left(\frac{8}{9}\right)^2}{-\frac{1}{9}\cdot 3\cdot\left(\frac{i}{3}-3i\right)}$$

$$= \frac{64\cdot 3}{-9\cdot 3\cdot i(1-9)} = \frac{8}{9i}$$

$$\text{Rez}(f,0) = \lim_{z\to 0}\left[z^2\cdot\frac{(z^2+1)^2}{z^2(3z^2-10iz-3)}\right]^1$$

$$= \lim_{z\to 0}\frac{2(z^2+1)\cdot 2z(3z^2-10iz-3)-(z^2+1)^2\cdot(6z-10i)}{(3z^2-10iz-3)^2}$$

$$= \frac{0-1\cdot(-10i)}{(0-0-3)^2} = \frac{10i}{9}$$

$$I = -\frac{1}{2}\cdot 2\pi i\left(\frac{10i}{9}+\frac{8i}{9}\right) = -\pi i\cdot\frac{18i}{9} = 2\pi$$

ex $\int_0^{2\pi} \frac{\cos x \cdot \cos(nx)}{13-12\cos x}\,dx \overset{\text{not}}{=\!=} I_1 \qquad n \in \mathbb{N}^*$

$\Rightarrow$

$$I_2 = \int_0^{2\pi} \frac{\cos x \sin(nx)}{13-12\cos x}\,dx$$

$$\Rightarrow I_1 + iI_2 = \int_0^{2\pi} \frac{\cos x \cdot e^{inx}}{13-12\cos x}\,dx$$

$z = e^{ix} \longrightarrow dz = ie^{ix}dx \Rightarrow dx = \frac{dz}{iz}$

$x \in [0, 2\pi] \longrightarrow |z| = 1$

$$\cos x = \frac{e^{ix}+e^{-ix}}{2} = \frac{z+z^{-1}}{2} = \frac{z^2+1}{2z}$$

$$\Rightarrow I_1 + iI_2 = \int_{|z|=1} \frac{\frac{z^2+1}{2z} \cdot z^{n-1}}{13-12\cdot\frac{z^2+1}{2z}} \cdot \frac{dz}{iz} = \int_{|z|=1} \frac{(z^2+1)\cdot z^{n-1}}{26z-12(z^2+1)} \frac{dz}{i}$$

$$I_1 + iI_2 = -\frac{1}{2i} \int_{|z|=1} \frac{(z^2+1)\,z^{n-1}}{6z^2-13z+6}\,dz$$

$6z^2-13z+6=0$

$\Delta = 169 - 4\cdot 6\cdot 6 = 169-144 = 25 \longrightarrow z_{1,2} = \frac{13 \pm 5}{2\cdot 6}$

$z = \frac{2}{3} \in \text{Int}(|z|=1)$ pol de ordinul 1

$z = \frac{3}{2} \notin \text{Int}(|z|=1)$

$$I_1 + iI_2 = -\frac{1}{2i} \cdot 2\pi i \operatorname{Rez}\left(f, \frac{2}{3}\right)$$

$$\operatorname{Rez}\left(f, \frac{2}{3}\right) = \lim_{z \to \frac{2}{3}} \left(z - \frac{2}{3}\right) \cdot \frac{z^{n-1}(z^2+1)}{6\left(z-\frac{2}{3}\right)\left(z-\frac{3}{2}\right)} = \frac{\left(\frac{2}{3}\right)^{n-1} \cdot \left(\frac{4}{9}+1\right)}{6\left(\frac{2}{3}-\frac{3}{2}\right)}$$

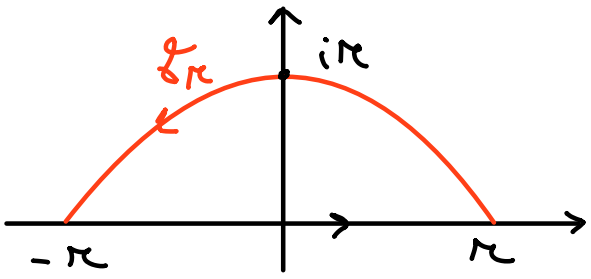
$$= \frac{\left(\frac{2}{3}\right)^{n-1} \cdot \frac{13}{9}}{4-9} = \frac{\left(\frac{2}{3}\right)^{n-1} \cdot \frac{13}{9}}{-5} = \left(\frac{2}{3}\right)^{n-1} \cdot \frac{13}{9} \cdot \left(-\frac{1}{5}\right) = \frac{-2^{n-1} \cdot 13}{5 \cdot 3^{n+1}}$$

$$I_1 + i I_2 = \frac{-\pi 2^{n-1} \cdot 13}{5 \cdot 3^{n+1}} \Rightarrow I_1 = \frac{-\pi 2^{n-1} \cdot 13}{5 \cdot 3^{n+1}}$$

$$I_2 = 0$$

Tipul II $\int_{-\infty}^{\infty} R(x)\,dx$, unde R este o funcție rațională fără poli pe axa reală astfel încât $\lim_{|x| \to \infty} x \cdot R(x) = 0$

Se consideră funcția complexă $R(z)$ și conturul $\gamma = \gamma_r \cup [-r, r]$, unde $r > 0$ ales astfel încât toți polii α_k, cu $\operatorname{Im} \alpha_k > 0$ să se afle în interiorul lui γ



$$\int_{\gamma} R(z)\,dz \overset{\text{th. rez}}{=} 2\pi i \sum_{\substack{k=1 \\ \operatorname{Im} k > 0}}^{n} \operatorname{Rez}(R(z), \alpha_k) \quad (*)$$

$$\int_{\gamma_r} R(z)\,dz + \int_{-r}^{r} R(x)\,dx$$

Lema Jordan

Fie f o funcție continuă pe vectorul $\theta_1 \le \theta \le \theta_2$ ($\theta = \arg z$, $z = e^{i\theta}$) a.î. $\lim\limits_{|z|\to\infty} zf(z) = 0$

atunci $\lim\limits_{r\to\infty} \int\limits_{\delta_r} f(z)\,dz = 0$

unde δ_r este un arc centrat în origine de rază r, situat în vectorul $\theta_1 \le \theta \le \theta_2$

În (*) trecem la limită după $r \to \infty$ $\Rightarrow$

$$\Rightarrow 2\pi i \sum_k \operatorname{Rez}(R(z), a_k) = \lim_{r\to\infty} \int_{\delta_r} R(z)\,dz + \int_{-\infty}^{\infty} R(x)\,dx$$

$\int\limits_{\delta_r} \to 0$ cf. Lema Jordan

(ex) $\int\limits_0^{\infty} \frac{a+x^2}{1+x^4}\,dx \quad a \in \mathbb{R}$

f e funcție pară $\longrightarrow \int\limits_0^{\infty} \frac{a+x^2}{1+x^4}\,dx = \frac{1}{2}\int\limits_{-\infty}^{\infty} \frac{a+x^2}{1+x^4}\,dx$ (ultima: $\int\limits_{-\infty}^{\infty}$ cf. Lema Jordan)

Considerăm $f(z) = \frac{a+z^2}{1+z^4}$

$z^4+1 = 0 \Rightarrow z^4 = -1 \Rightarrow z_k = \sqrt[4]{-1} = (-1)^{1/4} = e^{\frac{1}{4}\ln(-1)}$

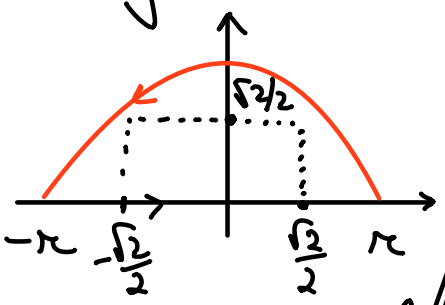
$$z_k = \left\{ e^{1/4\left[\ln|-1| + i(\arg(-1)+2k\pi)\right]},\ k = \overline{0,3} \right\}$$

$$z_k = \left\{ e^{i/4(\pi+2k\pi)},\ k = \overline{0,3} \right\} = \left\{ e^{i\pi/4};\ e^{i3\pi/4};\ e^{i5\pi/4};\ e^{i7\pi/4} \right\}$$

$z_k = \left\{ \frac{\sqrt2}{2} + i\frac{\sqrt2}{2}; -\frac{\sqrt2}{2} + i\frac{\sqrt2}{2}; -\frac{\sqrt2}{2} - i\frac{\sqrt2}{2}; \frac{\sqrt2}{2} - i\frac{\sqrt2}{2} \right\}$

poli de ordinul 1

Alegem $r > 1$ astfel incât $\gamma = \gamma_r \cup [-r; r]$



$$\int_\gamma f(z)dz \overset{\text{th. Rez.}}{=} 2\pi i \left(\operatorname{Rez}\left(f, \frac{\sqrt2}{2} + i\frac{\sqrt2}{2}\right) + \operatorname{Rez}\left(f, -\frac{\sqrt2}{2} + i\frac{\sqrt2}{2}\right) \right)$$

$$\int_\gamma f(z)dz = \int_{\gamma_r} f(z)dz + \int_{-r}^{r} f(x)dx \quad (1)$$

$$\lim_{|z|\to\infty} z \cdot f(z) = \lim_{|z|\to\infty} \frac{z(a+z^2)}{1+z^4} = 0 \overset{\text{Lema Jordan}}{\Longrightarrow} \int_{\gamma_2} f(z)dz \xrightarrow[r\to\infty]{} 0$$

În (1) fac $r \to \infty \Rightarrow K = \int_{-\infty}^{\infty} f(x)dx = ?$

$$\operatorname{Rez}\left(f, \underbrace{\frac{\sqrt2}{2} + i\frac{\sqrt2}{2}}_{z_0}\right) = \lim_{z\to z_0} (z - z_0) \frac{a+z^2}{(z-z_0)(z-z_1)(z-z_2)(z-z_3)} =$$

$$\left(e^{i\pi/4}\right)^2 = e^{i\pi/2}$$

$$= \frac{a + \left(\frac{\sqrt2}{2} + i\frac{\sqrt2}{2}\right)^2}{\sqrt2 \cdot (\sqrt2 + i\sqrt2) i\sqrt2} = \frac{a + e^{i\pi/2}}{2i\sqrt2(1+i)} = \frac{a+i}{2i\sqrt2(1+i)}$$

$$\operatorname{Rez}\left(f, \underbrace{-\frac{\sqrt2}{2} + i\frac{\sqrt2}{2}}_{z_1}\right) = \lim_{z\to z_1} (z - z_1) \frac{a+z^2}{(z-z_0)(z-z_1)(z-z_2)(z-z_3)} =$$

$$\left(e^{i3\pi/4}\right)^2$$

$$= \frac{a + \left(-\frac{\sqrt2}{2} + i\frac{\sqrt2}{2}\right)^2}{-\sqrt2 \cdot i\sqrt2 \cdot (-\sqrt2 + i\sqrt2)} = \frac{a + e^{i3\pi/2}}{2i\sqrt2(1-i)} = \frac{a-i}{2i\sqrt2(1-i)}$$

$$I = 2\pi i \left(\frac{a+i}{2i\sqrt{2}(1+i)} + \frac{a-i}{2i\sqrt{2}(1-i)} \right) = \frac{2\pi i}{2i\sqrt{2}} \left(\frac{a+i}{1+i} + \frac{a-i}{1-i} \right)$$

$$I = \frac{\pi}{\sqrt{2}} \cdot \frac{(a+i)(1-i)+(a-i)(1+i)}{1-i^2} = \frac{\pi}{\sqrt{2}} \cdot \frac{a-ai+i-i^2+a+ai-i-i^2}{2}$$

$$I = \frac{\pi}{\sqrt{2}} \cdot \frac{2a+2}{2} = \frac{\pi}{\sqrt{2}}(a+1)$$

$$\int_0^{\infty} \frac{a+x^2}{1+x^4}\,dx = \frac{1}{2} \cdot \frac{\pi}{\sqrt{2}}(a+1) = \frac{\pi(a+1)}{2\sqrt{2}}$$

Tipul III $\int_{-\infty}^{\infty} f(x) \cdot e^{ix}\,dx$, unde f este o funcție olomorfă pe $y>0$, eventual cu excepția unui nr. finit de puncte

Presupunem f nu are puncte singulare pe axa reală

Se parcurg pașii de la tipul II

Lema Jordan

Fie f o funcție continuă pe sectorul $\theta_1 \le \theta \le \theta_2$ situată în semiplanul $y>0$ și $\lim_{|z| \to \infty} f(z) = 0$

Atunci $\int_{\delta_r} f(z) e^{iz}\,dz \xrightarrow[r \to \infty]{} 0$, unde δ_r este un arc centrat în origine de rază r, situat în sectorul $\theta_1 \le \theta \le \theta_2$

ex $\int_0^\infty \frac{x \sin(ax)}{(x^2+b^2)(x^2+c^2)}\,dx \overset{\text{Not}}{=} I$ $\qquad a,b,c>0;\ c>b$

$f(x)$

f pară $\Rightarrow I = \frac{1}{2}\int_{-\infty}^{\infty} f(x)\,dx$

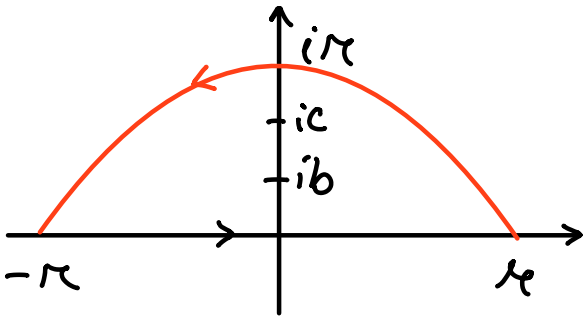
I_1

$I_2 = \int_{-\infty}^{\infty} \frac{x\cos(ax)}{(x^2+b^2)(x^2+c^2)}\,dx$

$\Rightarrow I_2 + iI_1 = \int_{-\infty}^{\infty} \frac{x e^{iax}}{(x^2+b^2)(x^2+c^2)}\,dx$

Fie $g(z) = \frac{z\cdot e^{iaz}}{(z^2+b^2)(z^2+c^2)}$

$\pm bi; \pm ci$ poli de ordinul 1



Alegem $R > c$ astfel încât

$\gamma = \gamma_R \cup [-R; R]$

$\int_\gamma g(z)\,dz \overset{\text{th. Rez.}}{=} 2\pi i\left(\text{Rez}(g,bi) + \text{Rez}(g,ci)\right)$

$\int_\gamma g(z)\,dz = \int_{\gamma_R} g(z)\,dz + \int_{-R}^{R} g(x)\,dx$

Trecem la limită (pt $R \to \infty$)

$\lim_{R\to\infty} \int_{\gamma_R} g(z)\,dz + \int_{-\infty}^{\infty} g(x)\,dx = 2\pi i\left(\text{Rez}(g,bi) + \text{Rez}(g,ci)\right)$ $\Rightarrow$

$$\lim_{|z|\to\infty} \frac{z}{(z^2+b^2)(z^2+c^2)} = 0 \overset{\text{Lema Jordan}}{\Longrightarrow} \lim_{R\to\infty} \int_{g_R} g(z)\,dz = 0$$

$$\Rightarrow J_2 + iJ_1 = \int_{-\infty}^{\infty} g(x)\,dx = 2\pi i \left(\operatorname{Rez}(g,bi) + \operatorname{Rez}(g,ci)\right)$$

$$\operatorname{Rez}(g,bi) = \lim_{z\to bi} (z-bi)\frac{z\,e^{iaz}}{(z-bi)(z+bi)(z^2+c^2)} = \frac{bi\cdot e^{-ab}}{2bi(c^2-b^2)}$$

$$\operatorname{Rez}(g,ci) = \lim_{z\to ci} (z-ci)\frac{z\,e^{iaz}}{(z^2+b^2)(z-ci)(z+ci)} = \frac{ci\cdot e^{-ac}}{2ci(b^2-c^2)}$$

$$J_2 + iJ_1 = 2\pi i \frac{1}{2(b^2-c^2)}\left(e^{-ac}-e^{-ab}\right) \Rightarrow J_2 = 0$$

$$\Rightarrow J_1 = \frac{\pi}{b^2-c^2}\left(e^{-ac}-e^{-ab}\right)$$

ex $\displaystyle\int_0^{\infty} \frac{x^2\sin^2 x + \cos^2 x}{(x^2+1)^2}\,dx \overset{\text{Not}}{=} J$

$$J = \frac{1}{2}\int_{-\infty}^{\infty} \frac{x^2\sin^2 x + \cos^2 x}{(x^2+1)^2}\,dx = \frac{1}{2}\int_{-\infty}^{\infty} \frac{x^2\,\frac{1-\cos 2x}{2} + \frac{1+\cos 2x}{2}}{(x^2+1)^2}\,dx$$

$$J = \frac{1}{4}\int_{-\infty}^{\infty} \frac{x^2(1-\cos x)+1+\cos 2x}{(x^2+1)^2}\,dx = \frac{1}{4}\int_{-\infty}^{\infty} \frac{x^2+1+\cos 2x(1-x^2)}{(x^2+1)^2}\,dx$$

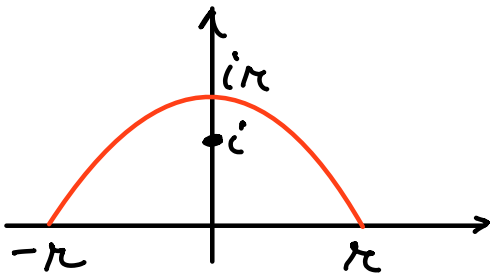
$$J = \frac{1}{4}\left(\int_{-\infty}^{\infty} \frac{1}{x^2+1}\,dx + \int_{-\infty}^{\infty} \frac{\cos 2x(1-x^2)}{(x^2+1)^2}\,dx\right)$$

$$\arctg x\Big|_{-\infty}^{\infty} = \pi \qquad \overset{\text{Not}}{=} J_1$$

Fie $I_2 = \int_{-\infty}^{\infty} \frac{\sin 2x (1-x^2)}{(x^2+1)^2} dx$ $\not\Rightarrow$ $I_1 + i I_2 = \int_{-\infty}^{\infty} \frac{(1-x^2)e^{2ix}}{(x^2+1)^2} dx$

Fie $f(z) = \frac{(1-z^2)e^{2iz}}{(1+z^2)^2}$

$1+z^2=0 \Rightarrow \pm i$ poli de ordinul 2



Alegem $r>1$ astfel încât $\gamma = \gamma_r \cup [-r; r]$

$$\int_{\gamma_r} f(z)dz + \int_{-r}^{r} f(x)dx = \int_{\gamma} f(z)dz \overset{\text{th. Rez.}}{=} 2\pi i \operatorname{Rez}(f, i)$$

Trecem la limită (pt. $r \to \infty$)

$$\lim_{r\to\infty} \int_{\gamma_r} f(z)dz + \int_{-\infty}^{\infty} f(x)dx = 2\pi i \operatorname{Rez}(f,i) \Rightarrow$$

$\lim_{|z|\to\infty} \frac{1-z^2}{(1+z^2)^2} = 0 \overset{\text{Lema Jordan}}{\Rightarrow} \lim_{r\to\infty} \int_{\gamma_r} f(z)dz = 0$

$$\Rightarrow I_1 + i I_2 = \int_{-\infty}^{\infty} \frac{(1-x^2)e^{2ix}}{(x^2+1)^2} dx = \int_{-\infty}^{\infty} f(x)dx = 2\pi i \operatorname{Rez}(f,i)$$

$$\operatorname{Rez}(f,i) = \lim_{z\to i} \left[\cancel{(z-i)^2} \frac{(1-z^2)e^{2iz}}{(z+i)^2 \cancel{(z-i)^2}} \right]'$$

$$\operatorname{Rez}(f,i) = \lim_{z\to i} \frac{[-2ze^{2iz} + (1-z^2)\cdot e^{2iz}\cdot 2i](z+i)^2 - (1-z^2)e^{2iz}\cdot 2}{(z+i)^3}$$

$$\operatorname{Rez}(f,i) = \frac{[-2ie^{-2} + (1-(-1))\cdot e^{-2}\cdot 2i]2i - (1-(-1))e^{-2}\cdot 2}{(2i)^3}$$

$$\operatorname{Rez}(f,i) = \frac{(-2ie^{-2} + 4ie^{-2})2i - 4e^{-2}}{-8i} = \frac{2ie^{-2}\cdot 2i - 4e^{-2}}{-8i}$$

$$\operatorname{Rez}(f,i) = \frac{-8e^{-2}}{-8i} = \frac{e^{-2}}{i}$$

$$J_1 + iJ_2 = \int_{-\infty}^{\infty} \frac{(1-x^2)e^{2ix}}{(x^2+1)^2}\,dx = 2\pi i\frac{e^{-2}}{i} = 2\pi e^{-2} \Rightarrow J_1 = 2\pi e^{-2}$$
$$J_2 = 0$$

$$\Rightarrow J = \int_0^{\infty} \frac{x^2\sin^2 x + \cos^2 x}{(x^2+1)^2}\,dx$$

$$J = \frac{1}{4}\left(\int_{-\infty}^{\infty} \frac{1}{x^2+1}\,dx + \int_{-\infty}^{\infty} \frac{\cos 2x(1-x^2)}{(x^2+1)^2}\,dx\right)$$

$$J = \frac{1}{4}\left(\operatorname{arctg}\Big|_{-\infty}^{\infty} + 2\pi e^{-2}\right) = \frac{1}{4}\left(\pi + 2\pi e^{-2}\right)$$